

Jacob Lamadrid

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EDUCATION

Georgia Institute of Technology

Master of Science in Analytics

Atlanta, GA

Aug. 2025 – Present

- Analytical Tools Track

- OMSA, Part-time

University of California, San Diego

Bachelor of Science in Cognitive Science (Machine Learning), Minor in Data Science

San Diego, CA

Sept. 2020 – June 2024

- 3.5 GPA; 3.75 Major GPA
- Honoree of Provost Honors
- 2024 Cognitive Science Summer Scholar (\$6,000 Research Award)
- Instructional Assistant (Python, Data Science)

EXPERIENCE

Research Fellow

UC San Diego Natural Computation Lab

July 2024 – Sept. 2024

San Diego, CA

- Conducted independent research on audio reconstruction from EEG using CNNs, LDMs, and DCGANs
- Processed and visualized EEG/MEG/audio data with signal processing methods for model input
- Designed data pipelines and experiments for multimodal signal processing and model evaluation
- Translated theoretical ML work into functional experiments and prototypes using PyTorch and custom datasets

Application Developer & Research Data Associate

UC San Diego Health

June 2022 – Sept. 2024

San Diego, CA

- Developed a cross-platform app using Flutter and Firestore to log developmental trends in neurodevelopmental clinical studies
- Built end-to-end data processing tools for signal and behavioral data to streamline statistical modeling workflows
- Conducted algorithmic analysis on eye-tracking and psychological assessment data to optimize data quality and structure
- Collaborated across diverse research staff to enter and manage expansive datasets and ensure high-quality outputs

Software Development Intern

NASA Langley Research Center

June 2023 – Sept. 2023

Hampton, VA

- Built data pipelines to migrate legacy datasets from the NASA ASDC to the TOLNet online repository used in atmospheric research
- Developed user interfaces for ongoing data migration projects
- Executed comprehensive data quality analytics tasks on the outcome of the migration for reporting to multiple research teams

PROJECTS

Audio Texture Synthesis | *Python, PyTorch, CNN, Signal Processing*

2023

- Generated realistic audio textures by training CNNs on spectrograms for audio extension tasks

MRI Tumor Classification | *Python, Scikit-learn, TensorFlow*

2024

- Achieved 96%+ accuracy in MRI tumor classification using PCA+MLP/KNN, comparable to CNN methods

WaveMAP Clustering | *Python, UMAP, Clustering, Signal Processing*

2024

- Clustered neuronal ERP signals using unsupervised ML and signal processing methods

TECHNICAL SKILLS

Languages: Python, Matlab, R, SQL, JavaScript

Frameworks/Tools: PyTorch, TensorFlow, Scikit-learn, Jupyter, Spark, Flutter, Firestore, Git, Bitbucket, React

Domains: Machine Learning, Deep Learning, Signal Processing, Computational Neuroscience, Neural Interfaces